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Roll No

EC-4002-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Signals and Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) With the help of examples, explain the following properties of signals. 7
 - i) Time shifting
 - ii) Time scaling
 - iii) Time reversal
- b) State sampling theorem. Define Nyquist rate. What is the Nyquist rate for the signal $x(t) = \sin c^2(100zt)$? 7
2. a) Which of the following systems are BIBO stable? 7
 - i) $y[n] = a^{x[n]}$
 - ii) $y[n] = |x[n]|$
- b) Which of the following systems are non linear? 7
 - i) $y(t) = 2x(t-1) - 3x(t-2) + x(t-3)$
 - ii) $y(t) = 2x(t) + 3.6$

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3. a) What is convolution integral? State the distributive, commutative and associative property of convolution. 7
b) The unit impulse response of a linear time invariant system is the unit step function $u(t)$. For $t > 0$, find the response of the system to an excitation $e^{-at} u(t)$ for $a > 0$. 7
4. a) Find the z-transform and ROC of the following sequences 7
i) $x(n) = u(n) - u(n-3)$
ii) $x(n) = \{1, 2, 3, 2\}$
iii) $x(n) = \{1, 2, -1, 2, 3\}$
b) State and prove the scaling and time shifting properties of z - transform. 7
5. a) What is the necessary and sufficient condition on the impulse response for
i) Causality
ii) Stability 4
b) The impulse response $h(n)$ of a linear time invariant system is given by $h(n) = u(n+3) + u(n-2) - 2u(n-7)$
Is the system stable? Is the system causal? 4
c) Find the convolution of 6
 $x_1(n) = \{1, -2, 3, 1\}$
 $x_2(n) = \{2, -3, 2\}$
6. a) Write any two properties of Discrete Time Fourier Transform (DTFT) and prove them. 7
b) Find the Fourier series of the following discrete-time signal. 7

$$x(n) = 1 + \sin\left(\frac{2\pi}{5}n\right) + 3\cos\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n + \frac{\pi}{2}\right)$$

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7. a) Obtain the direct form-I realization for the system described by the difference equation

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n) + 2x(n-1) \quad 7$$

- b) Obtain parallel form realization of the discrete time system described by the difference equation 7

$$y(n) - \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$$

8. Write short notes on any two topics. 14

- a) ROC of finite duration sequence
- b) Applications of DTFT
- c) Classification of signals
- d) State space analysis
